BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

DIGITAL CIRCUITS (2023 (REGULAR))

Exam Date: Tuesday, 3 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2023 (Regular)	Digital Electronics	100403	End Semester Examination - 2023

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven Question only: [14]

1. Draw Truth table of JK flip flop.
2. Convert the Decimal number (108.025) to their binary equivalent.
3. Draw the truth table and logic circuit of Half-adder.
4. Write the difference between combinational & Sequential circuit.
5. Draw timing diagram of SR flip-flop.
6. Find 2's complement of 1011011.
7. Number of 2:1 mux requires designing 256:1 mux is
8. Add hexadecimal number 2ABC & 98F2.
9. Discuss Universal gates.
10. What is the use of K-map?

Q.2 Solve this question : [14]

1. Implement the following function using only NAND gate $F(A, B, C) = \Sigma m(0, 1, 2, 3, 7)$.

Q.3 Solve both questions : [7+7=14]

1. Realize XNOR logic function using NAND gate only.
2. Simplify $Y = ABC + AB\overline{C} + A\overline{B}C$

Q.4 Solve this question : [14]

1. A logic circuit has four inputs A,B,C,D and output Y. Y=1 when A & B are both 1, subjected to the condition that C and D are both low or both high. Design the logic circuit.

Q.5 Solve this question : **[14]**

1. Explain Master-slave flip flop. What are race around condition?. How it can be circumvented with the help of Master-slave flip flop.

Q.6 Solve both questions : **[7+7=14]**

1. Design 8 to 3 line Encoder circuit.
2. Implement NAND gate using TTL logic family.

Q.7 Solve this question : **[14]**

1. Summarize the design procedure for a synchronous sequential circuit.

Q.8 Solve both questions : **[7+7=14]**

1. Explain the working of R-2R Ladder DAC.
2. Design 3-bit binary counter using T flip-flop.

Q.9 Write short notes on the following: **[14]**

1. (a) Binary Parallel Adder
2. (b) Digital IC logic families